

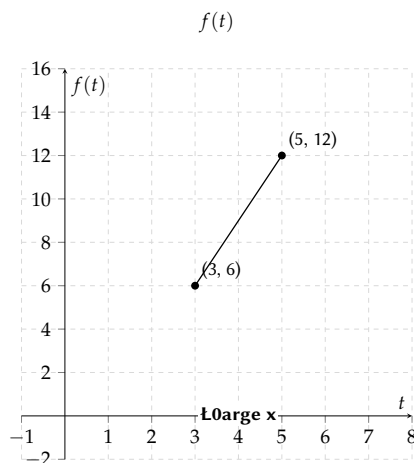
CPE 381 CLASSWORK 04: DISCRETE AND CONTINUOUS REPRESENTATIONS

Student Name (as in Canvas): _____

A Number: _____

Points: 20 Duration: 15 minutes Due: September 02, 2026 (In-Class)

1. Consider a signal $f(t)$ where $f(3) = 6$, and $f(5) = 12$. What will be the value of signal at $t = 4$, i.e. $f(4)$ if you happen to employ linear interpolation. **(10 Points)**

**Answer**

2. What are some non-ideal scenarios you may encounter while dealing with real-world signals? (5 Points)

Answer

3. Consider a MATLAB code snippet for the time vector

```
t = 0:0.03:10;
```

- (a) What is the sampling time period T_s ? (2.5 Points)
(b) What is the final time ? (2.5 Points)

Answer